SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I S.KAMALLI(192524423)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

Annexure A

Analytical Problem Solving

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

Scenario: A grayscale image of size 1024 × 1024 is captured using a surveillance camera under poor lighting conditions. The image is corrupted by salt-and-pepper noise with a noise density of 0.05. This means that some pixels are randomly changed to either black or white intensity values. The image has to be preprocessed before applying an edge detection algorithm. Since edge detection is sensitive to abrupt intensity variations, the selected filter should remove noise effectively while preserving important edge details.

Answer the following:

a. Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.
b. Explain why the selected filter performs better than an averaging filter for impulse noise removal.
c. If a 3 × 3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.
d. Estimate the total number of noisy pixels present in the original image.
e. Estimate the expected number of noisy pixels remaining after median filtering.
f. Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

Question 2: Discretization of a Continuous Image

Scenario : An analog grayscale image has a smooth sinusoidal intensity variation along the horizontal direction. The intensity changes continuously with respect to the spatial coordinate x, while it remains constant along the y-direction. The image is defined over a spatial range of 0 to 255 in both directions. This continuous image has to be converted into a digital image for storage and further processing in a computer vision system.

Answer the following:

a. Explain how the continuous image can be converted into a digital image using sampling and quantization.
b. Identify the spatial variation pattern of the image in the x-direction.
c. Determine the minimum sampling requirement needed to avoid aliasing in the xdirection.
d. Explain what will happen if the image is sampled below the required sampling rate.
e. Suggest a suitable 8-bit quantization scheme for storing the image.

f. Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

Question 3: Edge Detection in Thin Line Images

Scenario: A 512 × 512 grayscale image contains very thin vertical and horizontal white lines of 1-pixel thickness on a dark uniform background. Such images are commonly used in document analysis, engineering drawing interpretation, circuit diagram processing, and road-lane detection.

The objective is to detect the line boundaries accurately without losing thin edge information. Answer the following:

a. Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.
b. For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.
c. Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.
d. Discuss the role of Gaussian smoothing in the Canny edge detection process.
e. Explain how Gaussian smoothing may affect edge localization, especially for thin lines.
f. If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.

Question 4: Feature Extraction for Handwritten Digit Recognition

Scenario: A computer vision system is being developed for recognizing handwritten digits from scanned images. The handwritten digits vary in size, orientation, stroke thickness, and writing style. Therefore, robust feature extraction is required before classification. Answer the following:

a. Explain the working principles of SIFT, SURF, and ORB feature extraction methods.
b. Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.
c. If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.
d. If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.
e. Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.
f. Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.

Question 5: Morphological Preprocessing for Character Recognition

Scenario: A binary image of size 256 × 256 contains scanned handwritten characters. Due to poor scanning quality, the image contains small white noise dots in the background and small black gaps inside the character strokes.

Before applying character recognition, the image must be cleaned using morphological preprocessing techniques.

Answer the following:

a. Identify the morphological operation suitable for removing small white noise dots from the background.
b. Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.
c. Explain the difference between opening and closing operations using simple imageprocessing terminology.
d. If a 3 × 3 square structuring element is used, explain how it affects thin character strokes.
e. Discuss the risk of using a large structuring element during preprocessing.
f. Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection

a. Most suitable filtering technique

The Median Filter is the most suitable filter for removing salt-and-pepper (impulse) noise while preserving edges.

b. Why median filter is better than averaging filter

Median Filter

Replaces the center pixel with the median value of neighboring pixels.

Removes isolated black and white noise effectively.

Preserves sharp edges because it does not average intensity values.

Averaging (Mean) Filter

Replaces the center pixel with the average of neighboring pixels.

Impulse noise significantly affects the average.

Produces blurred edges and reduces image sharpness.

Therefore, the median filter performs much better for impulse noise removal.

c. Probability that a noisy center pixel is corrected (3 × 3 median filter)

Assumptions

Noise density = 0.05 (5%)

Noise is randomly distributed.

At least 5 out of the 9 pixels in the window remain noise-free.

Probability that a neighboring pixel is not noisy:

$P(\text{good})=0.95$

The probability that the median is determined by the correct pixels is

$P=\sum_{k=5}^{9}\binom{9}{k}(0.95)^{k}(0.05)^{9-k}$

Since only 5% of pixels are noisy, this probability is extremely close to 1.

Estimated probability ≈ 99.99% (approximately 1).

d. Total noisy pixels in the original image

Image size: $1024\times1024=1{,}048{,}576$ pixels

Noise density = 5%

$1{,}04{,}576\times0.05=52{,}428.8$

≈ 52,429 noisy pixels

e. Expected noisy pixels remaining after median filtering

Since the median filter removes almost all isolated impulse noise,

Correction probability ≈ 99.99%.

Remaining noisy pixels

$52{,}429\times0.0001\approx5$

Approximately 5 noisy pixels remain (ideal estimate).

f. If the image contained Gaussian noise

The preprocessing strategy changes because Gaussian noise affects all pixels, not just isolated ones.

Suitable filters:

Gaussian filter

Wiener filter

Bilateral filter (better edge preservation)

The median filter is less effective for Gaussian noise because Gaussian noise is continuous rather than impulsive.

Question 2: Discretization of a Continuous Image

a. Conversion using sampling and quantization

Two steps are involved:

Sampling

Converts continuous spatial coordinates into discrete pixels.

Example: 256 × 256 or 512 × 512 pixels.

Quantization

Converts continuous intensity values into finite gray levels.

Example: 256 intensity levels (8-bit).

Thus,

Continuous Image → Sampling → Quantization → Digital Image

b. Spatial variation pattern

The intensity varies as a sinusoidal wave along the x-direction.

Along the y-direction, intensity remains constant.

c. Minimum sampling requirement

According to the Nyquist Sampling Theorem,

Sampling frequency must satisfy

$fs \geq 2fmax$

where

fs = sampling frequency

$fmax$= highest spatial frequency

Therefore, sample at at least twice the highest spatial frequency.

d. If sampled below the required rate

Aliasing occurs.

Effects:

False patterns appear.

Wrong frequencies are recorded.

Distorted image.

Loss of image information.

e. Suitable 8-bit quantization

Use

8-bit grayscale

256 intensity levels

Range:0−255

where

0 = black

255 = white

f. Effect of quantization

Image Quality

More bits → smoother image

Fewer bits → visible banding

Storage

Higher bits require more memory.

8-bit requires 1 byte per pixel.

Intensity Accuracy

Higher bit depth gives better representation of brightness levels.

Question 3: Edge Detection in Thin Line Images

a. Comparison of edge detectors

Feature	Sobel	Prewitt	Canny
Noise suppression	Good	Moderate	Excellent
Edge localization	Good	Fair	Excellent
Thin edge detection	Moderate	Moderate	Excellent
Computational cost	Medium	Low	High
Best for 1-pixel lines	Fair	Fair	Best

Therefore, Canny is the preferred detector.

b. Sobel gradient response

Standard Sobel operators

Horizontal:

Gx=[−101−202−101]

Vertical:

Gy=[−1−2−1000121]

For a horizontal edge, intensity changes vertically.

Hence,

Gx=0

Gy=±1020

(using intensity difference 255)

Gradient magnitude

G=02+1020 2=1020

Direction 90∘

c. Gradient magnitude and direction

Gradient magnitude

$G=Gx2+Gy2$

indicates edge strength.

Gradient direction

$\theta=\tan-1(GyGx)$

indicates edge orientation.

d. Role of Gaussian smoothing in Canny

Gaussian smoothing

Removes image noise.

Prevents false edge detection.

Produces stable gradients.

Improves overall edge quality.

e. Effect on thin lines

Advantages:

Reduces noise.

Disadvantages:

Slight edge blurring.

Thin 1-pixel lines may become weaker.

Edge localization shifts slightly.

f. If Canny threshold is reduced by half

Expected results:

More edges detected.

More weak edges retained.

Increased false edges due to noise.

Higher sensitivity.

Question 4: Feature Extraction for Handwritten Digit Recognition

a. Working principles

SIFT

Detects scale-space extrema using Difference of Gaussian.

Computes orientation.

Generates 128-dimensional descriptors.

SURF

Uses Hessian matrix.

Uses integral images.

Faster than SIFT.

ORB

Uses FAST keypoint detector.

Uses BRIEF descriptor with orientation.

Very fast and efficient.

b. Comparison

Property	SIFT	SURF	ORB
Scale invariant	Yes	Yes	Limited
Rotation invariant	Yes	Yes	Yes
Speed	Slow	Medium	Fast
Descriptor	128-D	64-D	Binary
Real-time	No	Moderate	Yes

c. Rotated by 45°

Reliable methods:

✔ SIFT

✔ SURF

✔ ORB (rotation invariant, though generally less robust than SIFT/SURF under challenging conditions)

d. Total keypoints

Average keypoints/image = 120

Images = 50

$120 \times 50 = 6000$

Total keypoints = 6000

e. Suitable dimensionality reducti

Use Principal Component Analysis (PCA).

Choose enough principal components to retain 95% variance.

f. Benefits of dimensionality reduction

Faster training.

Faster classification.

Lower memory usage.

Removes redundant features.

Reduces overfitting.

Improves classifier performance.

Question 5: Morphological Preprocessing for Character Recognition

a. Remove white noise dots

Use Opening.

Opening = Erosion followed by Dilation.

b. Fill small gaps

Use Closing.

Closing = Dilation followed by Erosion.

c. Difference between opening and closing

Opening	Closing
Removes small white objects	Fills small black holes
Smooths object boundaries	Connects nearby regions
Shrinks then restores objects	Expands then restores objects

d. Effect of a 3 × 3 square structuring element

Removes isolated noise.

Fills tiny gaps.

Slightly smooths boundaries.

May thin or slightly alter very fine character strokes.

e. Risk of using a large structuring element

A large structuring element can:

Remove important character details.

Merge nearby characters.

Distort stroke shapes.

Reduce recognition accuracy.

f. Complete preprocessing pipeline

Image acquisition (scan or camera capture)

Grayscale conversion (if needed)

Noise removal (median or Gaussian filtering, depending on noise type)

Binarization (e.g., Otsu thresholding)

Morphological opening to remove small white noise

Morphological closing to fill gaps and holes

Normalization (resize and align characters)

Thinning/Skeletonization (optional, for stroke-based analysis)

Feature extraction (e.g., SIFT, ORB, HOG, or zoning features)

Classification using a machine learning or deep learning model (e.g., SVM, k-NN, or CNN).